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A Novel Dual-Level Multi-Graph Convolutional Network for Parkinson's Disease Diagnosis

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***Abstract*—Parkinson's disease (PD) is a common and complex neurodegenerative disorder. Its early discovery based on brain network analysis is critical for conducting effective clinical intervention. However, most existing methods depend mainly on single-modality information due to the lack of multimodal data. In this work, we utilize dual-level *i.e.* individual data and population data collected from fMRI and DTI imaging modalities respectively, to design a knowledge-driven dual-level multi-graph convolutional network (K-DMGCN) for early detection of PD. On the one side, we construct a knowledge-guided functional graph based on neuroanatomical priors, a data-guided functional graph via self-attention, and a structural graph from structural connections, aiming to mine brain connectivity relationships at the individual level. On the other side, cross-modal similarity graphs are built by integrating phenotypic data with image features, aiming to find common pathological patterns across different patients through adaptive graph convolution. A Top-K pooling operation is introduced to screen critical brain regions to increase the model's interpretability. In addition, we design a structural-functional discrepancy loss function to reduce multimodal information redundancy. Experimental results on the dataset from the Affiliated Hospital of Qingdao University demonstrated that the proposed K-DMGCN achieved superior classification performance over state-of-the-art methods. It achieved an accuracy of 90.21% for PD classification. Furthermore, we found that the most PD-relevant brain regions can be identified through cross-modal collaborative learning, which may provide a fresh understanding to explore the pathological mechanisms of PD.**

Keywords- Brain disease diagnosis; graph neural networks; multimodal data fusion; phenotypic information.

I. INTRODUCTION

Parkinson's disease (PD) is a prevalent and intricate neurodegenerative disease of the central nervous system, which typically presents with resting tremor, bradykinesia, and rigidity. With age-related incidence growth, PD ranks among the most common neurodegenerative diseases globally[1]. Timely and precise diagnosis is crucial for intervening in its pathological progression. Multimodal neuroimaging techniques, including functional magnetic resonance imaging (fMRI)[2] and diffusion tensor imaging (DTI)[3], have gained traction in brain network analysis. Specifically, fMRI-derived BOLD signals excel at identifying dynamic disruptions in functional connectivity across brain regions. In contrast, the information presented by DTI images facilitates detecting structural connectivity damage through tracking white matter fibers. However, utilizing medical imaging data has limitations for mining the hidden interconnectivity between functional dynamics and structural stability[4]. More recently, it has been proved that clinical phenotypic data, e.g., age and gender, are significantly associated with the pathological progression of PD[5]. Hence, how to comprehensively leverage the complementary information of multiple modalities is significant for analyzing multi-dimensional brain networks.

Originally, traditional machine learning methods were introduced to explore information fusion in brain network analysis[6]. However, when dealing with high-dimensional heterogeneous brain network data, it is limited due to challenges such as feature selection bias and the curse of dimensionality. Fortunately, graph convolutional networks (GCNs) offer a paradigm for integrating multimodal information and phenotypic data owing to their powerful representation capabilities for non-Euclidean data structures[7]. Hence, GCNs become a competitive tool for cross-modal and cross-scale collaborative learning based on graph-structured modeling in brain network analysis. However, there are still primary limitations in brain network analysis using GCNs. First, the individual-level research usually focuses solely on functional brain networks and fails to effectively incorporate the

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information of structural brain networks[8]. Second, the population-level investigations pay more attention to common patterns, while overlooking specific features of individual subjects[9]. Moreover, although prior knowledge can be exploited to construct brain networks, it is hard to fully capture the dynamic and intricate structure-function coupling mechanisms among brain neurons, either from the individual level or the population level[10].

To address these challenges, we design a knowledge-driven dual-level multi-graph convolutional network (K-DMGCN) for effective identification of PD, through systematic integration of features from multiple perspectives. The main contributions of this work are as follows: 1) To comprehensively characterize the multidimensional features of brain networks, a graph model is constructed by integrating triple information, *i.e.* knowledge-driven functional feature, data-driven functional feature, and structural feature at the individual level. 2) A population graph is constructed involving phenotypic data and image features, which is helpful to capture pattern correlations among subjects at the population level. 3) A novel loss function is designed to strengthen the multimodal complementary learning and feature fusion of individual-population information.

The paper is organized as follows. The details of the designed model are presented in Section II. Section III describes the experiments and results. In the last section, concluding remarks are drawn, and future directions are pointed out.

II. METHOD

The whole pipeline of K-DMGCN is illustrated in Figure 1. Multi-graph convolution serves to collectively capture brain network features across individual and population tiers. At the individual level, a multi-graph structure is developed comprising a knowledge-driven functional graph, a data-driven functional graph, and a structural graph. For the population level, a dual-modal similarity graph is constructed by integrating phenotypic data with image features to capture disease-related patterns. Through this architecture, the model can learn local and global features from individual and population graphs respectively. Then, the critical biomarkers can be screened out via adaptive fusion and Top-K pooling. By introducing constraints of structural-functional discrepancy and multi-graph discrepancy losses, the model can leverage the complementary information from both prior knowledge and image data.

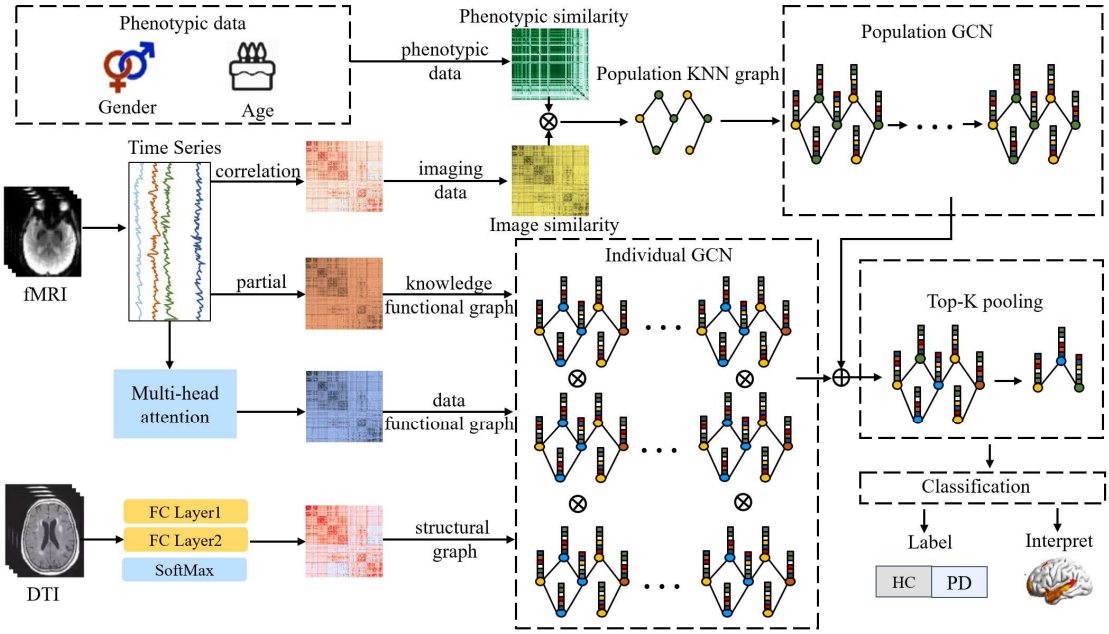

Figure 1. The Pipeline of the Proposed K-DMGCN.

A. Graph Construction of Individual Level

From the viewpoint of the individual level, three kinds of information sources are employed to construct the multi-graph. The first is a knowledge-driven functional graph A^{I_1}, which relies on AAL atlas priors for ROI nodes, with edges computed from BOLD signals to capture dynamic functional interactions. The second is a data-driven functional graph A^{I_2}, which is constructed by learning latent connectivity patterns from fMRI data via self-attention, without prior knowledge. The third is a structural graph A^{I_3}, which is constructed by features extracted from DTI images. For each subject, the fMRI feature matrix is defined as $T \in R^{N\times v\times l}$, Here, N denotes the sample count, v denotes the number of brain regions, while l signifies the length of the time series. The DTI feature matrix is formulated as $D \in R^{N\times v\times v}$. The calculations for obtaining the three graphs are as follows.

1) Calculation of A^{I_1}

The knowledge-driven functional graph A^{I_1} which relies on prior knowledge (AAL atlas) defines ROIs, while edge weights quantify dynamic functional interactions via BOLD signals.

To accurately evaluate the functional connectivity relationship between two brain regions, we first calculate the Pearson correlation coefficient:

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$$x_{mn} = \frac{\sum_{k=1}^{l} (t_{mk} - \bar{t}_m)(t_{nk} - \bar{t}_n)}{\sqrt{\sum_{k=1}^{l} (t_{mk} - \bar{t}_m)^2}\sqrt{\sum_{k=1}^{l} (t_{nk} - \bar{t}_n)^2}} \quad (1)$$

where t_{mk} represents the BOLD signal value of the m-th ROI at the k-th time point; $\bar{t}_m$ *and* $\bar{t}_n$ denote the means of the time series for the m-th and n-th brain regions, respectively.

Subsequently, the partial correlation matrix is derived from the calculated Pearson correlation coefficient, which is used as edge weights of A^{I_1}:

$$A_{mn}^{I_1} = \frac{x_{mn} - \sum_{k=1}^{l} (x_{mk} \cdot x_{nk})}{\sqrt{1 - \sum_{k=1}^{l} x_{mk}^2}\sqrt{1 - \sum_{k=1}^{l} x_{nk}^2}}, k \neq m, n \quad (2)$$

2) Calculation of A^{I_2}

The model directly extracts topological information from fMRI data without prior knowledge. A multi-head self-attention mechanism is utilized to dynamically seize inter-regional brain connections. First, the Query, Key, and Value matrices (Q_i, K_i, V_i) of the ith subject are calculated as follows:

$$Q_i, K_i, V_i = TW_i^q, TW_i^k, TW_i^v \quad (3)$$

where T denotes the time series matrix of fMRI images, W_i^q, W_i^k, W_i^v denote the learnable weight matrices. Then, the single-head attention is calculated as follows:

$$\text{head}_i = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d}}\right) V_i \quad (4)$$

where d denotes the attention dimension.

Then, the time-encoded features E are obtained through cascading operations of concatenating, weighting and Softmax as follows:

$$E = \text{Softmax}\left(\text{concat}\left(\text{head}_1, \dots, \text{head}_{i,\dots}\right) W_{\text{MLP}}\right) \quad (5)$$

where W_{MLP} represents the weight matrix of the multi-layer perceptron.

Finally, the fMRI-driven functional graph A^{I_2} is generated via the outer product of E as follows:

$$A^{I_2} = EE^T \quad (6)$$

3) Calculation of A^{I_3}

The structural data graph A^{I_3} is constructed based on the structural connectivity matrix D obtained from DTI images after preprocessing. The cascading operations of a fully connected network are expressed as follows:

$$A^{I_3} = \text{softmax}(W_2 \cdot \text{ReLU}(W_1 D + b_1) + b_2) \quad (7)$$

where W_1,b_1 and W_2,b_2 are the learnable parameters of the two layers respectively. The operation of Softmax normalizes the structural connectivity strength between brain regions row by row.

B. Graph Construction of Population Level

From a population perspective, every subject is mapped to a node, with weighted edges between nodes encoding pairwise subject similarities. Two kinds of information sources are utilized. The first is the phenotypic data which is employed to construct the phenotypic population graph A^{P_1}. The second is the image features of fMRI which are exploited to construct the imaging population graph A^{P_2} via K-nearest neighbor algorithm. Accordingly, two population graphs are constructed.

For the phenotypic graph, the similarity matrix is calculated as follows:

$$A_{ij}^{P_1} = \begin{cases} \alpha, & |P_{age}(i) - P_{age}(j)| < \beta, \\ & P_{gender}(i) = P_{gender}(j) \\ 1 - \alpha, & \text{otherwise} \end{cases} \quad (8)$$

where P_{age} and P_{gender} denote age and gender information respectively. α and β are hyperparameters which indicate 0.9 and 2 respectively.

For the imaging population graph, the imaging feature vector X_i is defined as the vectorized functional connectivity matrix generated by computing a symmetric 90×90 Pearson correlation matrix from preprocessed fMRI BOLD time series and flattening it into a vector $X_i \in R^{8100}$. A K-nearest neighbor algorithm with a Gaussian kernel is adopted:

$$A_{ij}^{P_2} = \begin{cases} \exp\left(-\gamma \|X_i - X_j\|_2\right), & X_j \epsilon \mathcal{N}_K(X_i), \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

here, γ denotes kernel width parameter, and $\mathcal{N}_K(X_i)$ denotes the set of K-nearest neighbors.

C. Information Fusion Through Multi-Graph Convolution

After the graphs are constructed, graph convolution is employed to integrate information across adjacent nodes and refresh node features. Initially, the graphs at the two levels undergo separate fusion processes as described below.

For the individual level, the information fusion formulation is as follows:

$$h_{I,i}^l = \sigma\left(\left(A^{I_1} h_{I,i}^{l-1}\right) \odot \left(A^{I_2} h_{I,i}^{l-1}\right) \odot \left(A^{I_3} h_{I,i}^{l-1}\right) W_I^l\right) \quad (10)$$

where $h_{I,i}^l$ represents the node feature vector of the i-th sample at layer l, with $h_{I,i}^0$ being the non-vectorized 90×90 FC matrix, $\odot$ represents element-wise multiplication, and W_I^l is a learnable weight matrix.

For the population level, the initial node features $H_P^0 = [X_1, X_2, \dots, X_N]^T \in R^{N \times 8100}$, the fusion operation is as follows:

$$H_P^l = \sigma\left((A^{P_1} \odot A^{P_2}) H_P^{l-1} W_P^l\right) \quad (11)$$

where H_P^l represents the node feature matrix of the population graph at the l-th layer, and W_P^l denotes learnable weight matrices.

Then, individual and population characteristics are fused as follows:

$$\hat{X} = \lambda_I H_I^L + \lambda_P H_P^L \quad (12)$$

where λ_I and λ_P are hyperparameters.

D. Key ROIs Selection and Disease Classification

To handle large node counts and high-dimensional features, we use Top-K pooling: nodes are ranked by importance scores, with the Top-K retained to enhance interpretability. The process is defined as:

$$\tilde{X} = TopK\left(\sigma\left(\hat{X} W_X\right), k\right) \quad (13)$$

where W_X denotes the learnable weight vector, and Top-K selects nodes with the highest scores. This retains key ROIs by polarizing node importance scores. These sparsified features retain the most discriminative brain imaging patterns for downstream classification.

The fused feature vectors $\tilde{X}$ preserved b y the pooling layer are fed into multiple fully connected layers for classification. Let Z^l denote the hidden features at the l-th layer, with the initial input being $Z^0 = \tilde{X}$. The computation at each layer is as follows:

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$$Z^l = \sigma(Z^{l-1}W_Z^l + b^l) \quad (14)$$

where W_Z^l and b^l denote the weight matrix and bias term, respectively.

After L layers of full connectivity, the predicted category probability distribution is generated by Softmax:

$$\hat{y} = Softmax(Z^L) \quad (15)$$

E. Loss Functions Design and Optimization

The cross-entropy loss function quantifies the divergence between the predicted probability distribution $\hat{y}$ and the ture label distribution y, with c representing the total number of classification categories:

$$\mathcal{L}_{ce} = -\frac{1}{N}\sum_{m=1}^{N}\sum_{c=1}^{C} y_{m,c} log(\hat{y}_{m,c}) \quad (16)$$

Intra and inter-class losses are employed to reduce intra-class variations while amplifying inter-class disparities, thereby improving class discriminability:

$$\mathcal{L}_{LDA} = \sum_{c\in C}\left(\frac{1}{|S_c|}\sum_{i\in S_c} \| A_{i,c} - \mu_c \|^2 - \sum_{c'\neq c} \| \mu_c - \mu_{c'} \|^2\right) \quad (17)$$

where S_c denotes the sample set of class c, and μ_c denotes the mean functional graph of class c.

The functional graph discrepancy loss enforces differentiation between the knowledge and functional data graphs:

$$\mathcal{L}_{\mathrm{fc}} = \frac{1}{n}\sum_{i=1}^{n} (A^{I_1} - A^{I_2})^2 \quad (18)$$

The structural-functional loss penalizes Frobenius norm differences between modality graphs to promote complementary learning:

$$\mathcal{L}_{\mathrm{sc}} = \frac{1}{N}\sum_{i=1}^{N} \left(\| A_i^{I_1} - A_i^{I_3} \|_F^2 + \| A_i^{I_2} - A_i^{I_3} \|_F^2\right) \quad (19)$$

The pooling regularization loss polarizes node importance scores for biomarker selection:

$$\mathcal{L}_{pool} = -\frac{1}{N}\sum_{n=1}^{N}\frac{1}{v}\left(\sum_{i=1}^{k} log\tilde{s}_{n,i} + \sum_{i=1}^{v-k} \log(1-\tilde{s}_{n,i+k})\right) \quad (20)$$

where $\tilde{s}_{n,i}$ denotes the importance score of the i-th node in sample n.

The final loss function is as follows:

$$\mathcal{L} = \mathcal{L}_{ce} + \lambda_1\mathcal{L}_{LDA} + \lambda_2\mathcal{L}_{fc} + \lambda_3\mathcal{L}_{sc} + \lambda_4\mathcal{L}_{pool} \quad (21)$$

III. Experiments and Results

A. Datasets

1) Data Acquisition: This study included 132 subjects from the Affiliated Hospital of Qingdao University, including 61 healthy controls (HCs) and 71 patients with PD. The rs-fMRI scanning parameters were as follows: repetition time (TR) = 2000 ms, echo time (TE) = 30 ms, flip angle (FA) = 90°, field of view (FOV) = 220×220 mm², and 240 time points. DTI scanning parameters comprised: TR = 8700 ms, TE = 90 ms, FA = 90°, FOV = 256×256 mm², resolution = 2.0×2.0×2.0 mm³, and 64 gradient directions.

2) Data Pre-Processing: The rs-fMRI data underwent preprocessing via the RESTplus V1.2 MATLAB toolbox[11]. This included segmentation, EPI template correction, realignment, and detrending to address confounding effects. The data was then divided into 240 time points, covering 90 ROIs as per the AAL parcellation scheme. In the preprocessing of DTI data, PANDA[12] software was utilized for distortion correction and anatomical region identification based on the AAL atlas, where the Fractional Anisotropy (FA) index served as an indicator of structural connectivity.

B. Experiment Settings

We trained and tested our model based on the PyTorch framework and ran it on the NVIDIA RTX A6000 48GB GPU. We randomly partitioned the datasets into training and test subsets via five-fold cross-validation. Adam was used as the optimizer. The initial learning rate and attenuation parameter settings were both 1e-4. The training lasted 400 epochs, with a batch size of 16 and a pooling rate of 0.7. To thoroughly assess the model's performance, a multi-dimensional evaluation metric system was adopted, including accuracy (ACC), sensitivity (SEN), specificity (SPE), F1 score (F1), and area under the curve (AUC). Analyzing these metrics collectively reflects the model's performance rigorously.

C. Results and Discussion

TABLE I. Performance Comparison of Different Methods on PD Datasets. The Values (%) are Denoted as Mean ± Std.

Method	ACC	SPE	SEN	F1	AUC
SVM[13]	62.41±7.76	71.58±8.22	67.73±9.97	70.73±8.66	62.81±7.37
MLP[14]	76.13±6.82	79.13±7.25	81.13±5.25	72.37±6.11	83.61±7.80
BrainGNN[15]	75.99±5.32	72.12±6.15	79.71±7.20	78.63±5.22	70.78±6.25
BrainTGL[8]	83.61±7.82	69.23±6.33	75.11±7.15	76.34±6.89	87.50±5.50
PopulationGCN[9]	76.50±3.64	75.18±2.20	69.87±4.20	73.82±3.12	73.27±3.04
GBDM[16]	83.93±4.10	79.66±5.36	71.00±7.21	64.58±5.32	64.58±5.32
MGCN[17]	82.77±4.30	85.41±6.51	81.76±5.26	78.52±4.22	86.83±7.15
Ours	**90.21±2.14**	**92.33±6.42**	**86.91±5.80**	**90.18±2.63**	**94.04±1.44**

To assess model efficacy, we conducted comparative analyses against both conventional machine learning approaches and state-of-the-art graph neural network architectures. Among all comparative methods in TABLE I, the proposed model achieved an accuracy of 90.21%, demonstrating superior performance compared to other methods. BrainGNN and BrainTGL utilize individual graphs derived from single-modality fMRI data, while GBDM and MGCN construct individual graphs using both structural and functional data. PopulationGCN utilizes population graphs. Our model employs dual-level multi-graph fusion, simultaneously preserving brain region

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topology and cross-sample commonality, and the performance has been significantly improved.

To identify brain regions associated with PD pathogenesis, we selected the top ten most significant regions based on node importance scores from the pooling layer. Using BrainNet Viewer[18], we visualized these PD-related key brain regions, as shown in Figure 2. The top ten significant regions were: postcentral gyrus (PoCG.L), inferior frontal gyrus, orbital part (ORBinf.L), rolandic operculum (ROL.L and ROL.R), postcentral gyrus (PoCG.R), caudate nucleus (CAU.R), superior temporal gyrus (STG.R), precentral gyrus (PreCG.L), lenticular nucleus pallidum (PAL.R), olfactory cortex (OLF.L). These findings are consistent with established neuroscience research[19, 20]. Notably, these regions align with PD's core pathological mechanisms, including sensorimotor integration deficits (PoCG/PreCG), dopaminergic dysfunction in basal ganglia circuits (CAU/PAL), and early non-motor manifestations (e.g., hyposmia linked to OLF.L). The bilateral abnormalities in rolandic operculum (ROL.L/R) further suggest widespread neural circuit disruption, demonstrating the model's effectiveness in identifying neurodegeneration signatures characteristic of PD.

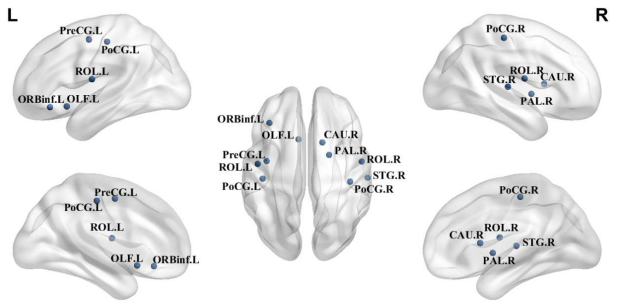

Figure 2. The top 10 significant brain ROIs of PD.

IV. Conclusion

In this work, a knowledge-driven dual-level multi-graph convolutional network (K-DMGCN) is designed to achieve precise diagnosis of PD. Multiple information sources, including fMRI images, DTI images, and phenotypic data, are collected and analyzed. The model implements feature extraction and information fusion at the individual and population levels through an end-to-end collaborative training and learning manner. Experimental results demonstrated that K-DMGCN achieved an accuracy of 90.21% in PD classification. Notably, the critical brain regions associated with PD pathogenesis identified via cross-modal collaborative learning offer potential neurobiological insights into the disease's pathological mechanisms. In future work, we plan to expand the dataset in size and incorporate other neuroimaging modalities, thereby enhancing both data volume and modality diversity. Moreover, we will optimize the model architecture to better handle complex, heterogeneous inputs, thereby enhancing its generalizability across diverse patient populations.

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